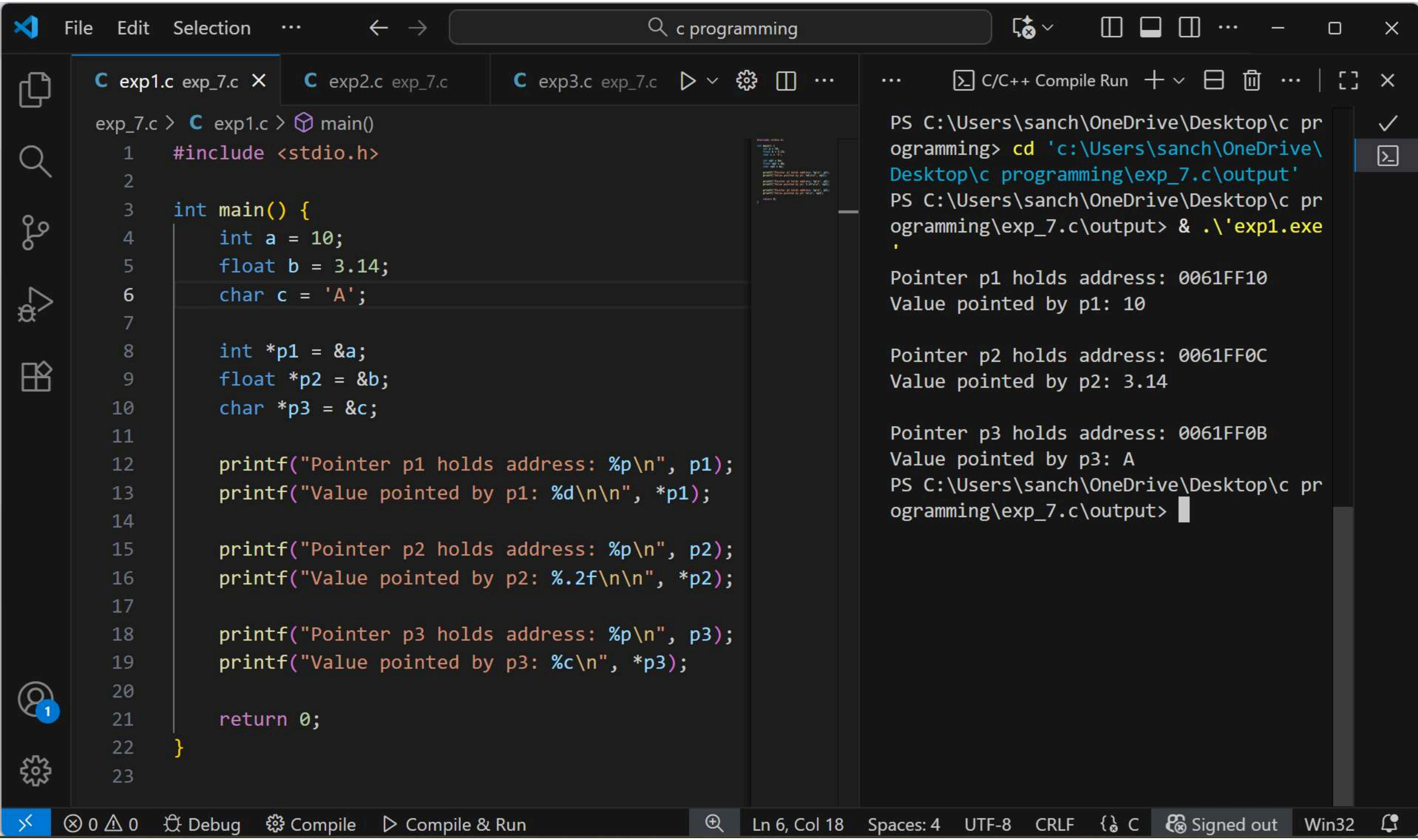
7.1

→ Declare different types of pointers (int, float, char) and initialize them with the addresses of variables. Print the value of both the pointers and the variables they point to

```
# include <stdio.h>
int main() {
  int a = 10;
float b = 3,14
  char c = 'A';

 int *p1 = &a;
float *p2 = &b;
char *p3 = &c;

printf("Pointer p1 hold address : %p\n", p1);
printf("Value pointed by p1 : %d\n\n", *p1);

printf("pointer p2 hold addres: %p\n", p2);
printf("Value pointed by p2: %.2f\n\n", *p2);

printf("Pointer p3 holds addres : %p\n", p3);
printf("Value pointed by p3 : %c\n", *p3);

return 0;
}
```


File Edit Selection ···
c programming
exp1.c exp_7.c
exp2.c exp_7.c
exp3.c exp_7.c
C/C++ Compile Run
exp_7.c > exp1.c > main()
#include <stdio.h>

int main() {
    int a = 10;
    float b = 3.14;
    char c = 'A';

    int *p1 = &a;
    float *p2 = &b;
    char *p3 = &c;

    printf("Pointer p1 holds address: %p\n", p1);
    printf("Value pointed by p1: %d\n\n", *p1);

    printf("Pointer p2 holds address: %p\n", p2);
    printf("Value pointed by p2: %.2f\n\n", *p2);

    printf("Pointer p3 holds address: %p\n", p3);
    printf("Value pointed by p3: %c\n", *p3);

    return 0;
}
PS C:\Users\sanch\OneDrive\Desktop\c programming> cd 'c:\Users\sanch\OneDrive\Desktop\c programming\exp_7.c\output'
PS C:\Users\sanch\OneDrive\Desktop\c programming\exp_7.c\output> & .\'exp1.exe'
Pointer p1 holds address: 0061FF10
Value pointed by p1: 10
Pointer p2 holds address: 0061FF0C
Value pointed by p2: 3.14
Pointer p3 holds address: 0061FF0B
Value pointed by p3: A
PS C:\Users\sanch\OneDrive\Desktop\c programming\exp_7.c\output>
0 0 Debug Compile Compile & Run
Ln 6, Col 18 Spaces: 4 UTF-8 CRLF C Signed out Win32

NAME

Exp - 7.2

→

Perform pointer arithmetic (increment & decrement) on pointers of different data access

→

```
#include <stdio.h>

int main() {
  int a = 10 ;
  float b = 20.5 ;
  char c = 'Z' ;

int * p1 = &a ;
float * p2 = &b ;
char * p3 = &c ;

printf ("The initial memory addresses stored in
  the pointers are as follow :\n");
printf ("The memory address stored in integer
  pointer p1 is %p \n", p1) ;
printf (" The meomory addresss stored in float
  pointer p2 is :%p \n", p2) ;
printf (" Ther memory address stored in character
  pointer p3 is %p\n", p3);
```

```
p1 ++ ;
p2 ++ ;
p3 ++ ;

printf ("\n After incrementing each pointer by one
        position: \n");
printf ("The new memory address stored in integer
  pointer p1 is : %p\n", p1);
printf ("The new memory address stored in float
  pointer p2 is %p\n" , p2 );
printf ("The new memory address stored in character
  pointer p3 is ; %p\n", p3 );

p1 -- ;
p2 -- ;
p3 -- ;

printf ("\n After decrementing each pointer back to its
  original position : \n");
printf ("The memory addrss stored in integer
  pointer p1 is now: %p\n", p1 );
printf ("The memory address stored in float pointer
  p2 is now: %p\n" , p2 );
printf ("The memory address stored in character
  pointer p3 is now : %p\n" , p3);
```

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```
printf ("\n The values stored at the memory location
 pointed to by each pointer are : \n");
printf (" The value pointed by p1 (integer pointer)
 %d \n * p1 );
printf (" The value pointed to by p2 ( float pointer)
 is : %.2f \n", *p2 );
printf (" The value pointed to by p3 ( character
 pointer) is : %.2f \n", *p3);

return 0;
}
```

```c
#include <stdio.h>

int main() {
    int a = 10;
    float b = 20.5;
    char c = 'Z';

    int *p1 = &a;
    float *p2 = &b;
    char *p3 = &c;

    printf("The initial memory addresses stored in the pointers are as follows:\n");
    printf("The memory address stored in integer pointer p1 is: %p\n", p1);
    printf("The memory address stored in float pointer p2 is: %p\n", p2);
    printf("The memory address stored in character pointer p3 is: %p\n", p3);

    p1++;
    p2++;
    p3++;

    printf("\nAfter incrementing each pointer by one position:\n");
    printf("The new memory address stored in integer pointer p1 is: %p\n", p1);
    printf("The new memory address stored in float pointer p2 is: %p\n", p2);
    printf("The new memory address stored in character pointer p3 is: %p\n", p3);

    p1--;
    p2--;
    p3--;

    printf("\nAfter decrementing each pointer back to its original position:\n");
    printf("The memory address stored in integer pointer p1 is now: %p\n", p1);
    printf("The memory address stored in float pointer p2 is now: %p\n", p2);
    printf("The memory address stored in character pointer p3 is now: %p\n", p3);

    printf("\nThe values stored at the memory locations pointed to by each pointer are:\n");
    printf("The value pointed to by p1 (integer pointer) is: %d\n", *p1);
    printf("The value pointed to by p2 (float pointer) is: %.2f\n", *p2);
    printf("The value pointed to by p3 (character pointer) is: %c\n", *p3);

    return 0;
}
```

```
PS C:\Users\sanch\OneDrive\Desktop\c programming> cd 'c:\Users\sanch\OneDrive\Desktop\c programming\exp_7.c\output'
PS C:\Users\sanch\OneDrive\Desktop\c programming\exp_7.c\output> & .\'exp2.exe'
The initial memory addresses stored in the pointers are as follows:
The memory address stored in integer pointer p1 is: 0061FF10
The memory address stored in float pointer p2 is: 0061FF0C
The memory address stored in character pointer p3 is: 0061FF0B

After incrementing each pointer by one position:
The new memory address stored in integer pointer p1 is: 0061FF14
The new memory address stored in float pointer p2 is: 0061FF10
The new memory address stored in character pointer p3 is: 0061FF0C

After decrementing each pointer back to its original position:
The memory address stored in integer pointer p1 is now: 0061FF10
The memory address stored in float pointer p2 is now: 0061FF0C
The memory address stored in character pointer p3 is now: 0061FF0B

The values stored at the memory locations pointed to by each pointer are:
The value pointed to by p1 (integer pointer) is: 10
The value pointed to by p2 (float pointer) is: 20.50
The value pointed to by p3 (character pointer) is: Z
PS C:\Users\sanch\OneDrive\Desktop\c programming\exp_7.c\output>
```

7.3

→

Write a function that accepts pointers as parameters. Pass variables by reference using pointers & modify their values within function

→

```
# include < stdio.h >

void modifyvalues (int * p1 , float *p2 , char * p3) {
  *p1 = *p1 + 5 ;
  *p2 = *p2 + 10;
  *p3 = 'A' ;
}

int main () {
    int a = 10 ;
  float b = 20.5 ;
  char c = 'Z' ;

printf (" The values of the variables before calling the
  function are : \n" ) ;
printf ("The value of integer variable a is %d \n", a);
printf (" The value of float variable b is %.2f \n", b);
printf ( " The values of character variable c is %c \n, c);
```

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```
modifyvalues (&a, &b, &c);

printf ("\n The values of the variables after calling
 the function are :\n");
printf ("The modified value of integer variable a is %d \n", a);
printf ("The modified value of float variable b is %.2f\n", b);
printf ("the modified value of character variable c is %c\n", c);

return 0;

}
```

```c
#include <stdio.h>

void modifyValues(int *p1, float *p2, char *p3) {
    *p1 = *p1 + 5;
    *p2 = *p2 + 10;
    *p3 = 'A';
}

int main() {
    int a = 10;
    float b = 20.5;
    char c = 'Z';

    printf("The values of the variables before calling the function are:\n");
    printf("The value of integer variable a is: %d\n", a);
    printf("The value of float variable b is: %.2f\n", b);
    printf("The value of character variable c is: %c\n", c);

    modifyValues(&a, &b, &c);

    printf("\nThe values of the variables after calling the function are:\n");
    printf("The modified value of integer variable a is: %d\n", a);
    printf("The modified value of float variable b is: %.2f\n", b);
    printf("The modified value of character variable c is: %c\n", c);

    return 0;
}
```

```
PS C:\Users\sanch\OneDrive\Desktop\c programming> cd 'c:\Users\sanch\OneDrive\Desktop\c programming\exp_7.c\output'
PS C:\Users\sanch\OneDrive\Desktop\c programming\exp_7.c\output> & .\'exp3.exe'
The values of the variables before calling the function are:
The value of integer variable a is: 10
The value of float variable b is: 20.50
The value of character variable c is: Z

The values of the variables after calling the function are:
The modified value of integer variable a is: 15
The modified value of float variable b is: 30.50
The modified value of character variable c is: A
PS C:\Users\sanch\OneDrive\Desktop\c programming\exp_7.c\output>
```